Red Hat Reference Architecture Series

Migrating Symantec Veritas Cluster Server to Red Hat Enterprise Linux HA

Clustered Apache Web Server
HP ProLiant DL580
Red Hat Enterprise Linux High Availability
Red Hat Enterprise Linux 6
Veritas Cluster Server

Version 1.1
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Migrating Symantec Veritas Cluster Server to Red Hat Enterprise Linux HA

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1 Executive Summary

The Red Hat® High Availability Add-On to Red Hat Enterprise Linux® allows IT staff to design a cluster of servers that provides its company with seamless performance in the event of component failure. This Add-On reduces downtime by insuring there is no single point of failure in a cluster and isolates unresponsive applications and nodes so they can't corrupt critical enterprise data.

This reference-architecture guide discusses how to migrate an application from a high-availability cluster running Symantec™ Veritas™ Cluster Server (VCS) to a cluster running Red Hat Enterprise Linux with the High Availability Add-On (Red Hat Enterprise Linux HA[1]).

A step-by-step process describes how to use configuration information from an existing VCS cluster to rapidly create a Red Hat Enterprise Linux HA cluster with similar functionality.

[1] In this guide, the term "Red Hat Enterprise Linux HA" is sometimes used in place of "Red Hat Enterprise Linux with the High Availability Add-On."

2 Introduction

In today's data centers, ensuring data integrity is as important as uptime in determining true availability. Red Hat Enterprise Linux offers many methods for ensuring high levels of data integrity for applications running in clustered environments, including situations when a cluster's active node fails.

Red Hat offers a High Availability Add-On to its enterprise-class Linux operating system. The Red Hat Enterprise Linux HA integrates with Red Hat Enterprise Linux and is updated as Red Hat Enterprise Linux is updated. The Red Hat Enterprise Linux HA provides the tools to create, configure, and maintain clusters of servers and virtual machines up to 16 nodes. Red Hat Enterprise Linux HA offers out-of-the-box support for multiple methods of ensuring data integrity in clusters, and includes support for hardware from APC®, WTI, Cisco®, Dell™, HP, and IBM®, among others.

This guide provides step-by-step instructions for migrating a highly available web server running on a VCS cluster with Red Hat Enterprise Linux 5.6 to one running on a Red Hat Enterprise Linux HA cluster with Red Hat Enterprise Linux 5.6, and to one running on Red Hat Enterprise Linux HA with Red Hat Enterprise Linux 6.1. See Figure 2-1 for the network layout of this two-node, clustered web service.

The cluster used here is a two-node, active/passive one; that is, although both servers (nodes) are running, the web server application runs on only one node (the active one). The second node (passive) stands ready to automatically run the web server should the active node fail to provide web service. Examples of failure include the active node crashing or hanging, a server component such as a NIC stopping working, the server losing its connection to the network, or the server being unable to use a critical resource, such as a SAN.

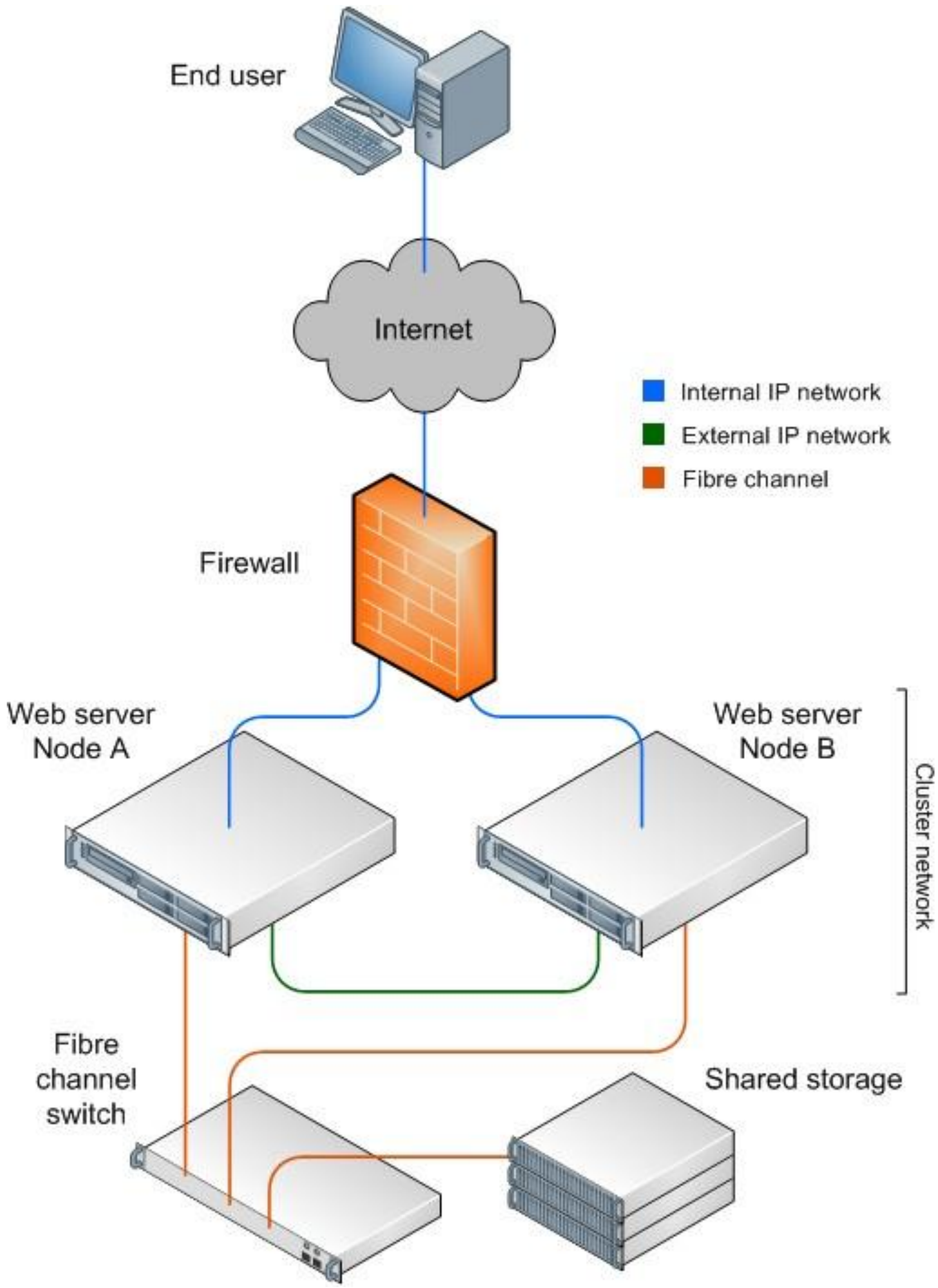


Figure 2-1: Network diagram for the clustered Apache web server

2.1 Audience

This document is for users of Symantec Veritas Cluster Server on Red Hat Enterprise Linux 5 who wish to migrate to Red Hat Enterprise Linux HA on Red Hat Enterprise Linux 5, version 5.6 or later; or Red Hat Enterprise Linux 6, version 6.1 or later. The procedures in this migration guide are suited to Linux system administrators who are familiar with the following activities:

- Linux system administration procedures, including kernel configuration
- Installation and configuration of shared storage networks, such as Fibre Channel SANs or iSCSI SANs

The guide assumes the reader is also acquainted with Symantec Veritas Cluster Server.

Details in the Red Hat documentation do not necessarily appear in this white paper. In addition to formal documentation from Red Hat, the upstream project for the Red Hat high-availability cluster solution maintains a rich wiki at https://fedorahosted.org/cluster/wiki/HomePage. However, take care when referring to documentation on the wiki, as formal Red Hat Enterprise Linux releases may not completely implement upstream features.

3 Red Hat Enterprise Linux with HA Add-On High Availability (HA) Features

Red Hat Enterprise Linux HA in both Red Hat Enterprise Linux 5.6 and Red Hat Enterprise Linux 6.1 and later provides tools to create, configure, and manage HA clusters with up to 16 nodes. (Please contact Red Hat sales or your support representative for details on support for larger clusters.)

Red Hat Enterprise Linux HA on Red Hat Enterprise Linux 5.6 offers several important improvements over previous version 5 releases, including the following:

- Ability to designate resources as critical or noncritical to a service
- Full support for SELinux in the enforcing state with a targeted policy

In addition, Red Hat Enterprise Linux HA on Red Hat Enterprise Linux 6.1 offers several important improvements over the 6.0 release, including the following:

- Ability to designate resources as critical or noncritical to a service
- An improved GUI management interface now offers a consistent set of screens
- Command-line cluster management tools are complete and simplified
- KVM guests as a cluster service for some fencing devices
- SNMP traps for many cluster events and conditions
- Full support for SELinux in the enforcing state with a targeted policy
- Restart-Disable failure policy, which disables a service if it cannot restart, instead of moving the service to a new node

3.1 Service and Resources

An HA service is a group of cluster resources configured into a coherent entity that provides specialized services to clients. The HA service comprises its resources, their properties, and their dependencies to other resources.

3.2 HA Services

Red Hat Enterprise Linux HA stores service configurations as XML trees in which the parent, sibling, and child relationships are manifest. Red Hat Enterprise Linux HA also provides out-of-the-box service definitions for several common Linux applications and system services, including the following:

- Apache™ web server
- MySQL®
- Oracle®
- PostgreSQL
- Apache Tomcat
- SAP®
- Open LDAP
- Samba
- HA Logical Volume Manager (HA LVM)
- NFS

- Application (Script)

3.3 Failover Domain

HA services run on only one node at a time in order to maintain data integrity. When a node providing a resource fails, it restarts on another node based on one of two configuration choices: either an explicitly stated order or, more simply, on any node in a specified set of nodes also known as a failover domain.

You can also specify what to do when the failed node returns to the cluster. It is inefficient for the service to return or failback to the original node, but in some cases, you may prefer that the service run on the original node as much as possible. In that case, you can configure the service case to failback to that node.

3.4 Cluster Management

Red Hat Enterprise Linux HA provides both web-based and command-line tools for managing clusters. This guide describes performing this migration to Red Hat Enterprise Linux 5.6 HA with the GUI, and to Red Hat Enterprise Linux 6.1 with command-line tools.

3.4.1 Cluster Management with Conga GUI Tools

Conga provides cluster, service, and resource management from a web browser. Conga runs either on a management server or directly on a cluster node. The Linux service that actually provides the GUI to the cluster is *luci*. You can connect via an SSL-enabled web browser to TCP port 8084 and log in with any user account on the server.

Management tasks available from the Conga GUI include the following:

- Clusters: Create and delete clusters. Manage an existing cluster.
- Nodes: Create and delete nodes, including installing cluster software. Add existing nodes to the cluster. Reboot nodes.
- Add and delete fence devices.
- Add and delete cluster resources. Configure resources.
- Create and delete service groups. Start, stop, and restart cluster services.
- Create and modify the cluster configuration, including properties of the Linux daemons that make up the cluster services, the intra-cluster or heartbeat network, universal logging options, and quorum disk configuration.

3.4.2 Cluster Management with Command Line Tools

You can manage Red Hat Enterprise Linux HA clusters with command-line tools. Red Hat Enterprise Linux 6.1 HA, in particular, offers straightforward, scriptable access to the cluster for creation, configuration, and management.

The cluster configuration resides in the file */etc/cluster/cluster.conf* in a structured XML format, and there is no longer a need to edit this file directly. The file format expresses relationships between components. For example, clusters have nodes; nodes are fenced; and fencing is executed by fence devices. Moreover, clusters run services; services run on sets of nodes; services respond to node failures; and services are composed of resources (web servers, IP address, application files, etc.).

4 HA Cluster Configuration Options

High Availability is difficult to measure because you cannot quantify it without experiencing failure—of the application, infrastructure, servers, or server components.

High Availability is a balance of design, cost, and maintainability. A cluster with a complex infrastructure design could provide higher availability, but often with greater cost in materials and maintenance. Red Hat's HA solution provides management tools that enable you to design, implement, and maintain clusters up to 16 nodes.

Currently, Red Hat Enterprise Linux fully supports only those clusters hosted entirely in one location. Red Hat does not formally support clusters spread across multiple physical locations. For more details, and to discuss multi-site clusters, please speak to your Red Hat sales or support representative.

4.1 Redundancy

The business requirements for the cluster solution usually provide a sufficient guide to the amount of redundancy needed in the service's design. A well implemented HA design uses redundant components and multiple data pathways to eliminate single points of failure. A component or IT service becomes a single point of failure when the availability and data-integrity requirements of a business application cannot be achieved when the component fails or operates in a degraded mode.

Many servers today are readily available with redundant components; for example, dual-controller RAID arrays, multiple bonded network channels, multiple paths between cluster members and storage, and uninterruptible power supply (UPS) systems.

Alternatively, you can set up a low-cost cluster to provide less availability than a no-single-point-of-failure cluster. For example, you can set up a cluster with a single-controller RAID array and only a single network channel. Certain low-cost alternatives, such as host RAID controllers, software RAID without cluster support, and multi-initiator parallel SCSI configurations are not compatible or appropriate for use as shared cluster storage.

4.2 I/O Fencing Considerations

To ensure data integrity, only one node can run a cluster service and modify cluster-service data at a time. Fence devices (hardware or software solutions that remotely power on/off, shutdown, and reboot cluster nodes) guarantee data integrity under all failure conditions. This prevents two nodes from simultaneously accessing the same data and corrupting it.

4.2.1 Quorum Rule

The cluster determines quorum by the communication among cluster nodes via the network. Optionally, you can determine quorum by a combination of network communication and a quorum disk. For network-based quorum, quorum consists of a simple majority (50 percent of the nodes + 1 extra). When configuring a quorum disk, quorum consists of user-specified conditions.

4.2.2 Using qdisk to Bolster Quorum

Often, the cluster functions better when some services run on one particular node. The services could run on the other nodes, but not as efficiently—for example, a node could have a faster processor or faster connection to the network or SAN. In these special cases, it is best to avoid fencing this node if it is isolated from the other nodes via the heartbeat network, but otherwise still able to perform its critical task. Ordinarily, the nodes in the other part of the cluster, a majority, would win a quorum vote, and the special node would be fenced.

For these special cases, Red Hat Enterprise Linux HA provides a disk-based method that does not rely on the heartbeat network to help determine quorum. This qdisk daemon allows the node to determine whether it can perform its critical tasks. If the isolated node is functioning properly, it wins and will provide that resource for the cluster. For reliability, Red Hat recommends that you use the qdisk only when the cluster uses power controllers to fence nodes.

You can also use qdisk to break ties in quorum votes, although a cluster with a well-designed network and fencing configuration should not need a tiebreaker. You might need a tiebreaker if you have a two-node configuration, either with the fence devices on a different network path than the path used for cluster communication, or where fencing is at the fabric level.

4.2.3 Multiple I/O Fences

With Red Hat Enterprise Linux HA, you can specify multiple fences for a node. If the first fence fails to disconnect the node, the other fences execute in turn until one works or none remain. With good redundant network paths and SAN paths, two distinct fences provide assurance that the node will disconnect. This migration example includes two fencing methods.

4.3 Networking Considerations

Nodes communicate with each other through the network and for many clusters with outside clients. At a minimum, a cluster uses a heartbeat network to determine membership and to exchange cluster data between nodes. It is a best practice to isolate the heartbeat network (minimally, at the VLAN level) from other physical or logical network connections to reduce resource contention, decrease latency, and usually to create an easier to maintain infrastructure. A redundant network design increases availability, but at the cost of additional network devices.

Red Hat recommends the use of pairs of bonded NICs (channel bonding) for the client and cluster networks to provide a good level of redundancy and reduce the number of cluster reformations. If possible, you should bond the ports from two separate NICs, and not ports from a multiport NIC, for each network.

5 Examining the Symantec Veritas Cluster Server (VCS) Configuration

Should local documentation of the original VCS cluster be incomplete, the VCS configuration files contain information that will aid setting up the Red Hat Enterprise Linux HA cluster. The most important items are the number and names of the server nodes, IP address, application information, shared file systems, and how the cluster responds to a failed component.

The cluster configuration is contained in a small number of files in the directory */etc/VRTSvcs/conf/config,* and the one most useful for the migration is *main.cf*. While these files do not use XML to denote configuration values and their relation to other objects, you can parse the file's format simply by viewing it. Alternatively, the Appendix, section 9.1, contains scripts that list the nodes, services, and resources in VCS clusters.

The cluster service entry (*ClusterService*) lists the nodes in the cluster (*SystemList*), and the name of the heartbeat NIC (*csgnic*). The IP address presented to the outside world (*webip*) for the web server is also specified.

```
group ClusterService (
    SystemList = { node1 = 0, node2 = 1 }
    …
    IP webip (
        Device = eth0
        Address = "10.41.5.40"
        NetMask = "255.255.248.0" )

    NIC csgnic ( Device = eth0 )
```

The *cvm* service entry (not shown) indicates that both nodes use VCS's shared file system.

The next service entry (*vrts_vea_cfs_int_cfsmount1*) provides the mount point (*MountPoint*) for the web server's application file system.

```
group vrts_vea_cfs_int_cfsmount1 (
    …
    CFSMount cfsmount1 (
        Critical = 0
        MountPoint = "/mount/vol1"
        BlockDevice = "/dev/vx/dsk/vxvol1/vol1"
```

The final entry describes the location of the Apache web server's binaries (*httpdDir*) and its configuration file (*ConfigFile*).

```
group webservice (
    …
    Apache Apache (
        httpdDir = "/usr/sbin"
        ConfigFile = "/etc/httpd/conf/httpd.conf" )
```

Note that other VCS configuration files in the directory further describe the services and how to manipulate them. Most of these, including Apache, are part of the standard VCS agent bundle. Additional configuration data for both supporting resources (e.g., DNS) and applications (Apache) are in the *types.cf* file.

6 I/O Fencing Details

Fencing provides data integrity by disconnecting (fencing) a misbehaving node from shared storage. Fencing methods include power fencing to power down the system, storage fencing to disable the FC port on the switches servicing the node, hardware-dependent methods that use an out-of-band channel to control the server, and SCSI-3 persistent reservations.

Note: Communication between the cluster and fenced device is usually network based. Be sure to add iptables firewall rules to allow communication between nodes and between the cluster and network devices.

6.1 SCSI-3 PR Fencing

The cluster's nodes register with SCSI devices supporting persistent reservations (PR) and obtain exclusive rights to read and write to the device. These reservations are stored with the device, and SCSI bus resets cannot remove them. You must deliberately remove the reservations. To fence a node, the host is removed from the device's registration group. Shared SCSI disks or SAN supporting PR are an excellent I/O fence: fencing operations are fast, split-brain situations become rare, and only one node can write to the device.

While SCSI-PR provides a high level of data protection, it cannot restart a dead or hung system. Either an administrator or a watchdog process must restart the server.

Red Hat Enterprise Linux HA in Red Hat Enterprise Linux 5.6 and 6.1 supports SCSI-3 PR fencing either with or without multipathing. The connection to a supported SAN device can be through FC or iSCSI.

6.2 HP iLO Fencing

HP iLO is an example of an out-of-band control channel for HP servers and blades. The iLO connection can be made through a serial port or network port (Red Hat Enterprise Linux HA uses the latter). iLO supports encrypted communication and LDAP authentication. An iLO fence can power on or power off the HP server.

6.3 SCSI-3 PR and iLO Fencing

In this migration example, you configure the Red Hat Enterprise Linux HA cluster attempt to fence a failed node with SCSI-3 PR, and to use iLO fencing if that method fails.

7 Conducting the Migration

This guide provides step-by-step instructions for migrating a highly available Apache web server from VCS on Red Hat Enterprise Linux 5.6 to Red Hat Enterprise Linux with HA Add-On on two HP ProLiant DL580 servers. The Red Hat Enterprise Linux HA cluster will use SCSI-3 PR fencing, backed by iLO fencing.

This guide assumes that business requirements for the migration dictate that the cluster's server platforms will be replaced with newer hardware. Accordingly, you have the option to run the VCS and Red Hat Enterprise Linux HA clusters in parallel, although that is not needed to perform the migration.

This guide assumes that the source cluster (VCS) is composed of two nodes: two HP ProLiant DL580 servers each with two NICs, and one FC HBA connected to a NetApp® storage array. One NIC connects the node to the outside, and the other is for the heartbeat network.

The NetApp SAN supports SCSI-3 PR, which makes it suitable for fencing. The SAN presents five LUNs to each VCS node: one LUN is an unshared boot and OS LUN, one is a shared LUN for the web server, and three are shared VCS-quorum LUNs. In total, the two-node cluster uses six LUNs. Each node runs Red Hat Enterprise Linux 5.6 x86_64 and VCS version 5.1.

The destination cluster (Red Hat Enterprise Linux HA cluster) is similarly provisioned: two newer HP ProLiant DL580 servers with two NICs and one FCX HBA each. One NIC connects to the client network, and the other to the heartbeat network. Red Hat recommends adding additional NICs as a straightforward way to eliminate single points of failure. You can also allocate additional HBA ports. This scenario does not implement that enhanced configuration.

This guide implements two fencing methods on the Red Hat Enterprise Linux HA cluster: SCSI-3 PR and, as a backup, HP iLO fencing.

In this configuration, the destination cluster uses the same NetApp array. It presents two LUNs to each node: one is an unshared boot and OS LUN, and one is a shared LUN for the web server. This two-node cluster uses three LUNs in total.

7.1 Planning

Even straightforward migrations such as this benefit from a planning process, if only to record the proposed steps of the procedure. But, in most cases, this process can help uncover long-forgotten configuration one-offs and custom scripts. While one person can perform this migration, other staff may be needed to allocate new resources or modify infrastructure configurations; for example, allocate new LUNs, change network firewall rules, reconfigure backups, etc.

7.1.1 Additional Considerations

- Estimate the length of the outage.
- Schedule the outage.
- Determine when and whom to contact regarding the outage.

- Obtain necessary subscriptions from Red Hat.
- Obtain new servers for the cluster.
- Determine whether any roles and responsibilities will change, and assign as needed.
- Update as much as possible procedures for the cluster.
- Define the criteria for a successful migration.
- Assign all tasks to the appropriate groups or individuals.

7.1.2 Migrating VCS Cluster and Resource Agent Configuration

- As early as possible, review of the configuration of the VCS cluster, and determine the cluster resources currently used against those actually needed.

7.1.3 Saving Customer-defined Software and External Scripts Configuration

- As early as possible, review the VCS nodes for custom scripts, software, license files, etc., that you must also migrate to the new cluster.

7.1.4 Backing up Application and Configuration Data

- When practical, schedule additional backups of the servers in the cluster, including the application's data and configuration files.

7.1.5 Back-out Plan

- While many back-out plans seem trivial ("reverse the steps"), consider writing out the steps in reverse order to assist you should backing out become necessary.

7.1.6 Post-migration Steps

- Perform all validation tests specified by the definition of a success migration.
- Shut down the VCS cluster.
- Ensure that you have updated processes and procedures.

7.2 Migrating the Cluster

This section provides step-by-step instructions for migrating a highly available Apache web server from VCS on Red Hat Enterprise Linux 5.6 to Red Hat Enterprise Linux HA. A complete description of how we configured the VCS cluster, the starting point for this migration, is in Section 10.3 in the Appendix.

Due to differences between major OS releases, this guide describes two migrations: first, to Red Hat Enterprise Linux HA running on Red Hat Enterprise Linux 5.6 and second, to Red Hat Enterprise Linux HA running on Red Hat Enterprise Linux 6.1. To illustrate two supported methods for creating and managing a Red Hat HA cluster, the migration to Red Hat Enterprise Linux 5.6 HA will use the Conga GUI, and the migration to Red Hat Enterprise Linux 6.1 HA will use the *ccs* utility from the command line. Of course, Red Hat supports the GUI and the command line for both Linux releases.

The instructions in the next three sections, 7.2.1-3, apply to both migrations. Section 7.2.4 applies only to the Red Hat Enterprise Linux 5.6 HA migration. Likewise, section 7.2.5 applies only to the Red Hat Enterprise Linux 6.1 HA migration. Both migrations complete the tasks in the final sections, 7.2.6-10.

7.2.1 Installing Red Hat Enterprise Linux on the New Servers

The server names are *HANode1* and *HANode2*

1. Verify the servers' network and disk configuration. The first interface on each server connects to the outside, and the second interface connects to the node's partner for intra-cluster communication. In addition, each HP ProLiant DL580 server has an iLO connection on an internal management network.

   HANode1's network interfaces and IP addresses
   - eth0: 10.41.5.127
   - eth1: 10.10.10.127
   - iLO: 192.168.72.240

   HANode2's network interfaces and IP addresses
   - eth0: 10.41.5.128
   - eth1: 10.10.10.128
   - iLO: 192.168.72.242

2. Each server has a 50GB boot from SAN disk for the Red Hat OS. In addition, all servers share one 10GB LUN for the application disk.
3. Perform steps 4-7 for each server.
4. Install and configure the base Red Hat Enterprise Linux operating system following standard methods.
5. SELinux runs in enforcing mode with targeted policy.
6. Configure two NICs with static IP addresses, as indicated in step 1 above. Note that you cannot use Network Manager with Red Hat Enterprise Linux HA.
7. Configure the iptables firewall to permit the following IP communications:

   - Outside access to the web server on TCP port 80
   - Fencing communications from the node to both servers' iLO on TCP port 22
   - Management access to the Conga GUI, if used, on TCP port 8084.
   - Cluster services' communication between nodes

   The IP ports used by the cluster services are found at http://docs.redhat.com/docs/en-US/Red_Hat_Enterprise_Linux/5/html/Cluster_Administration/s1-iptables-CA.html for release 5.6 and at http://docs.redhat.com/docs/en-US/Red_Hat_Enterprise_Linux/6/html/Cluster_Administration/s1-iptables-CA.html for release 6.1.

8. Configure and start the NTP time server. A consistent time source simplifies many administrative tasks, especially interpreting log messages.

7.2.2 Creating the Application's Shared File System

The new servers have new disk partitions and new Red Hat Enterprise Linux OSs. You cannot use the SCFS-formatted VCS LUN in the new cluster without VCS software. Instead, create a new shared LUN with an *ext3* file system, and copy the web server's files to it.

1. If not using multiple paths to the SAN, use *dmesg* and */proc/partitions* to identify the shared LUN; in our case, */dev/sda* is the boot/OS LUN and */dev/sdb* is the shared LUN.

```
[root@node1 /]# dmesg | egrep 'NETAPP|Attached scsi disk'
  Vendor: NETAPP    Model: LUN               Rev: 0.2
sd 0:0:0:0: Attached scsi disk sda
  Vendor: NETAPP    Model: LUN               Rev: 0.2
sd 0:0:0:1: Attached scsi disk sdb
```

2. Create one 20GB partition for the web server's file system with *parted*:

```
# parted /dev/sdb mkpart primary 0 20480
```

3. Initialize the partition for LVM, create one volume group named *web01*, create one logical volume named *weblv*, and create an *ext3* file system:

```
# pvcreate /dev/sdb1
# vgcreate web01 /dev/sdb1
# lvcreate –L 10G –n weblv web01
# mkfs.ext3 /dev/mapper/web01-weblv
```

4. Create a mount for the web server's file system on each node, as an example, using *ssh* in a simple bash loop:

```
# for host in HANode1 HANode2; do
        ssh $host mkdir –p /mnt/web
done
```

   Note: Do not add the file system to */etc/fstab* because the cluster software will mount it on the nodes as needed. See below.

5. In order to make this logical volume available as a shared resource to the cluster, we must modify LVM's configuration to ignore this volume unless it is tagged with the host's name.
   a. Log onto the first node as root, and edit the file */etc/lvm/lvm.conf*
   b. Go to the activation stanza, and the line *volume_list = ["VolGroup00", "@HANode1]*, where "VolGroup00" is the volume group for the root file filesystem.

```
activation {

# July 1, 2011
# The cluster uses logical volume "weblv" on volume group "web01" as an HA LVM resource.
# Accordingly, we add the volume group for the root filesystem.
#
    volume_list = [ "VolGroup00",  "@node2" ]
#
```

   c. Check the new lvm.conf for typographical errors, by typing *lvdisplay*. There should be no errors.

6. Update the node's initrd ramdisk to include this change:

a. For Red Hat Enterprise Linux 5.6, type:

```
# new-kernel-pkg --mkinitrd --initrdfile=/boot/initrd-Cluster-`uname -r`.img \
        --install `uname -r`
```

b. For Red Hat Enterprise Linux 6.1, type

```
# dracut -H -f /boot/initramfs-$(uname -r).img $(uname -r)
```

7. Reboot the server by typing *shutdown –r now*
8. Repeat steps 5-7 for the second node *HANode2*, replacing *“@HANode1”* with *“@HANode2”* in the new line in the file lvm.conf.

7.2.3 Installing Application Software: Apache Web Server

End users will connect to the Apache web server on TCP port 80 and IP address 10.41.5.106.

1. On each Red Hat Enterprise Linux HA node, install and configure Apache web server:

```
# yum groupinstall “Web Server”
# chkconfig httpd off
```

2. Change the web document root directory and network interface the server listens on. Edit the file */etc/httpd/conf/httpd.conf*, and replace the line

```
Listen 80
```

with

```
Listen 10.41.5.106:80
```

and the line

```
DocumentRoot "/var/ www/html"
```

with

```
DocumentRoot "/mnt/web/ "
```

At this point, the instructions split into two paths: the first path describes the migration to Red Hat Enterprise Linux 5.6 HA using the Conga GUI, and the second the migration to Red Hat Enterprise Linux 6.1 using the *ccs* command.

7.2.4 Migrating Via the Conga GUI to Red Hat Enterprise Linux 5.6 HA

For this migration, we will use only one fencing device, iLO, for the Apache web service.

7.2.4.1 Installing Red Hat HA Software

1. Log onto the first server, *HANode1*, as root.
2. Install the HA base software and HA management tools from these two groups of RPMs:

```
# yum groupinstall “High Availability”
# yum groupinstall “High Availability Management”
```

3. Start the secure intra-cluster communication service, ricci:

```
# service ricci start
```

chkconfig ricci on

4. Assign a password to the ricci user account by typing *passwd ricci*.

7.2.4.2 Creating the two-Node Cluster with Conga

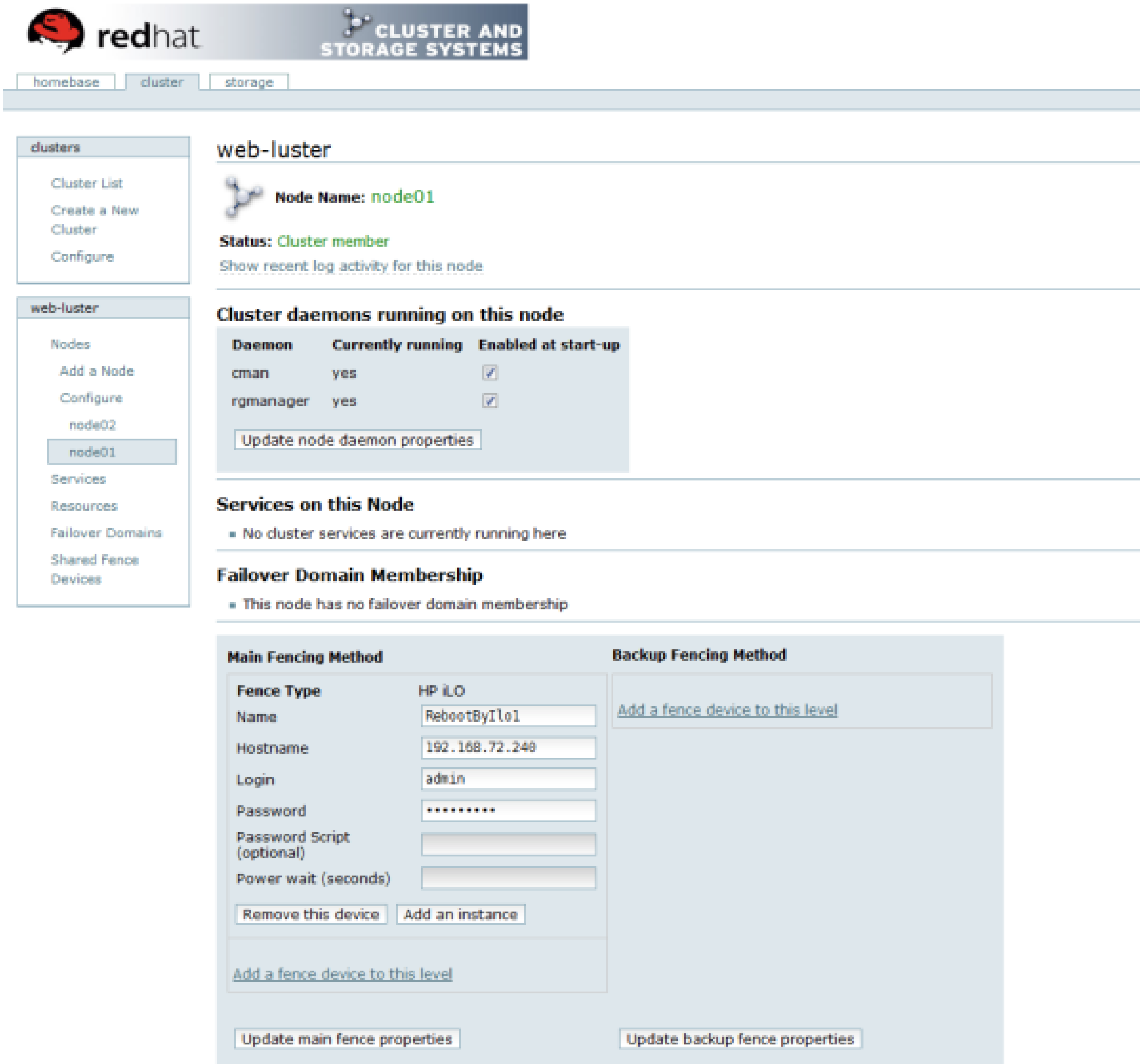
1. Log onto the first server, *HANode1*, as root.
2. Initialize the GUI, and set the password for the Conga administrator by typing *luci_admin init*
3. Restart the GUI service, *luci*, by typing *service luci restart*.
4. On a host with access to HANode1, start a web browser and open the URL https://HANode1:8084/.
5. Log into Conga using the admin account and the password created in step 2.
6. Select the cluster tab, and click Create a New Cluster.
7. At the Create a New Cluster screen, enter the cluster name web-*cluster*, and the two nodes' hostnames (not aliases).
8. Enter the password for the ricci account, and select the box Check if node passwords are identical.
9. Delete the box for a third node by clicking the trash icon to the far right.
10. (Optional) Check cluster communications by clicking View SSL fingerprints.
11. Click Submit to create the cluster with two nodes.

7.2.4.3 Configuring I/O Fencing

The cluster design calls for one fencing method for each node: HP iLO.

1. Add the iLO fence to the first node. Select the configuration page for node *HANode1*, click Add a fence device to this level under the Main fencing method. From the new pull-down menu, scroll to *HP iLO*. Enter *RebootByIlo1* for the method's name, *192.168.72.240* for the hostname, *admin* for the Login, and enter the Password for the iLO account. Click

Update main fence properties, and click OK in the Update this node's fence configuration pop-up.



2. Repeat step 1 for the second node, but name its iLO fence *RebootByIlo2* and change its hostname to *192.168.72.242*.

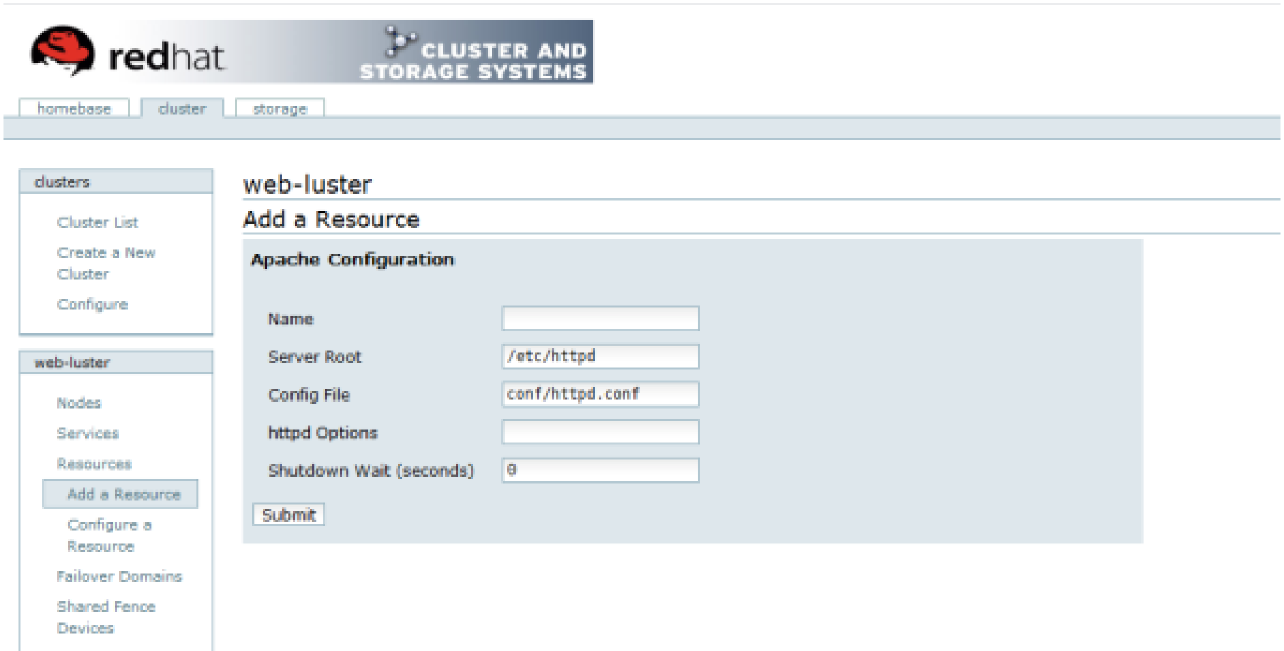
7.2.4.4 Configuring Failover Domains

For this two-node cluster, configuring failover domains is unnecessary, as the default failover policy suffices: if the active node or a critical component fails, activate the passive node, switch the shared LUN and virtual IP resources to the new active node, and start the web server.

7.2.4.5 Configuring the HA Web Server

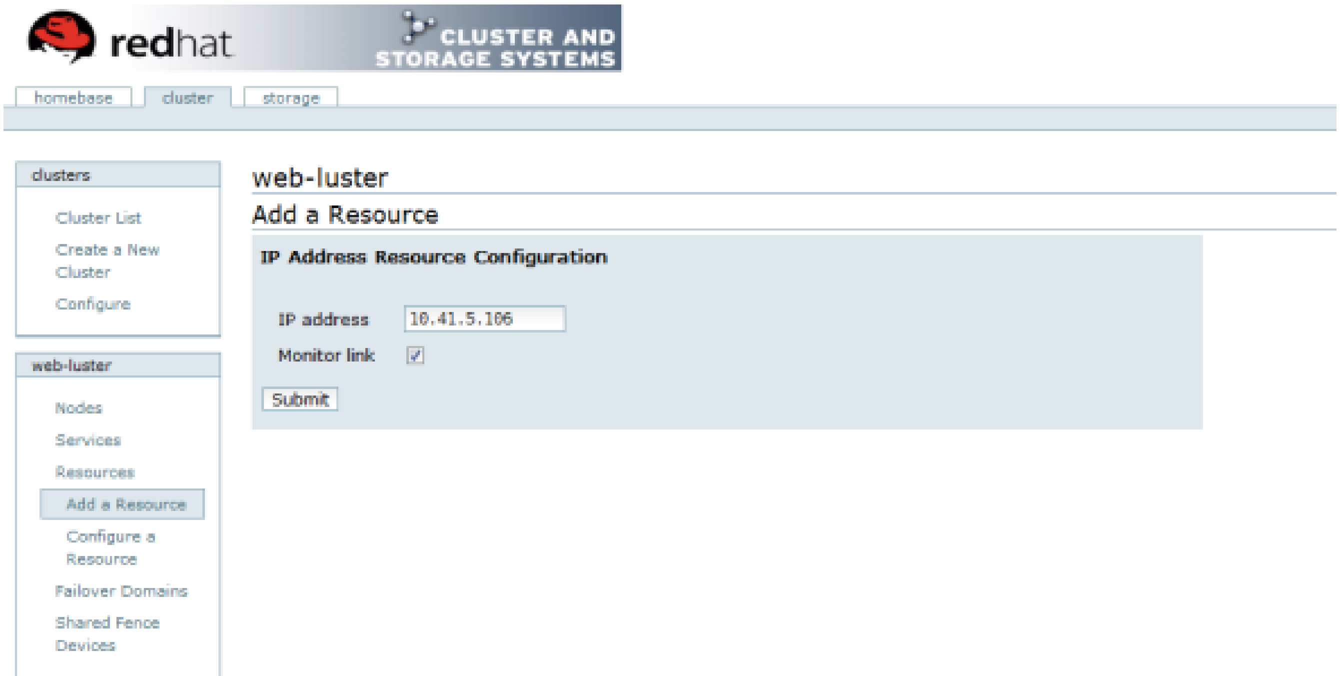
The clustered, high-availability Apache web server is a service requiring four cluster resources: the Apache software, the IP address it is bound to, the shared logical volume, and the latter's file system containing its document files. These resources are all part of Red Hat Enterprise Linux HA's default package.

1. Create the Apache web server resource, named *web*. Select the cluster from the menu on the left, press Resources, and then Add a resource. From the dropdown menu Select a Resource, scroll to Apache.

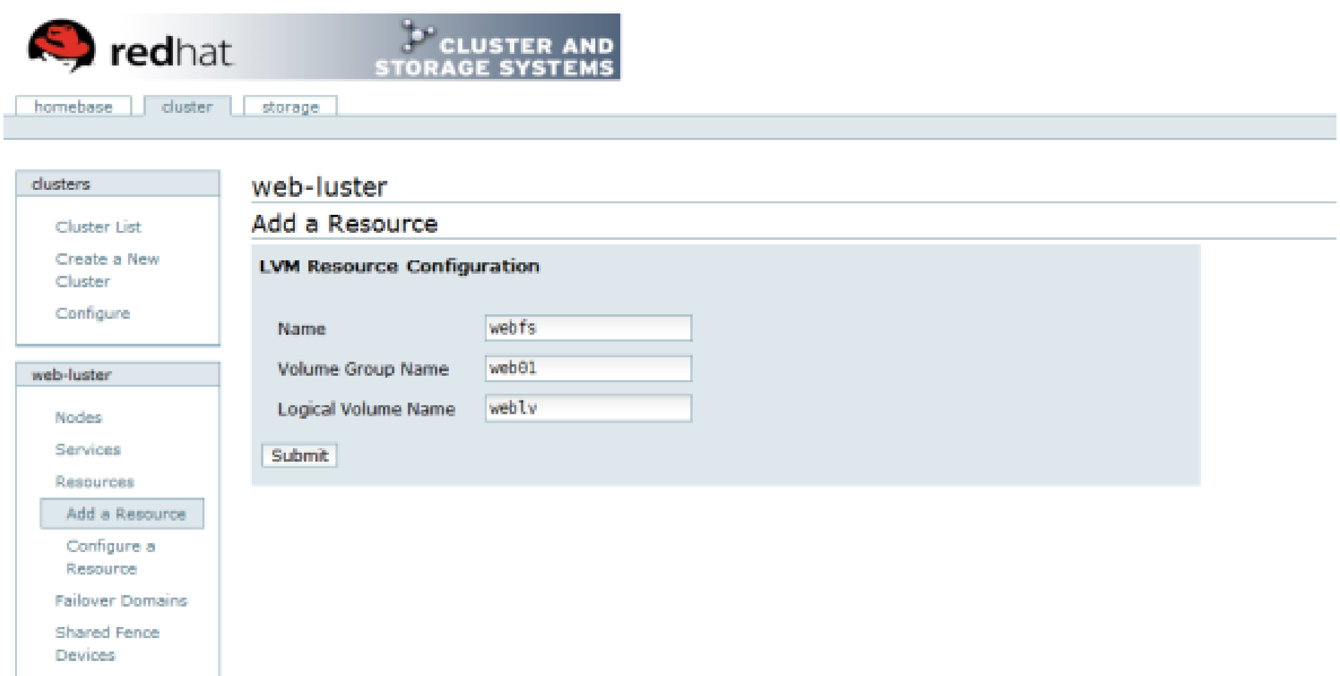


Enter *web* as the resource name. Accept the default values for *Server Root* and *Config File*, /etc/httpd and config/httpd.conf, respectively. Click Submit, and click OK in the Update this resource pop-up.

2. Create the web server's virtual IP address as a resource, and have the link monitored for failure. From the cluster's Resource menu, press Add a resource. From the Select a Resource drop-down menu, scroll to IP Address. Enter *10.41.5.106* for the IP address, and check the Monitor link box. Click Submit, and click OK in the Update this resource pop-up.

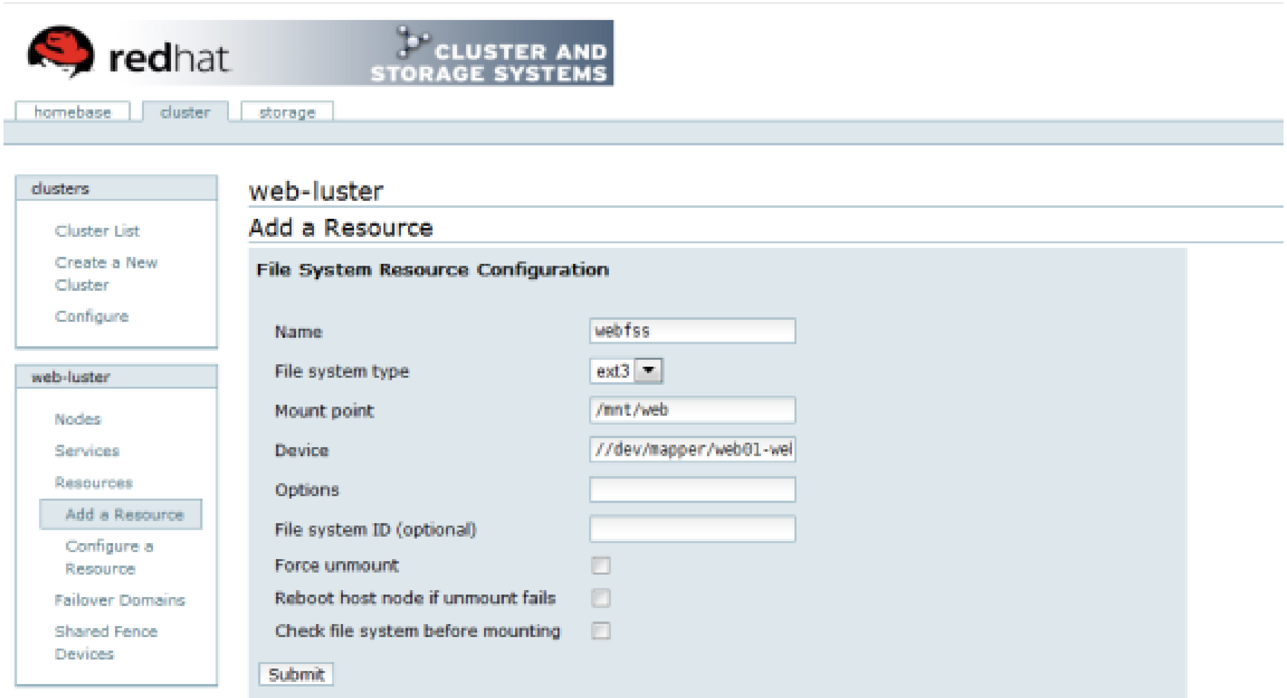


3. Create the shared logical volume resource, named webfs, for the logical volume created in section 7.2.2. From the cluster's Resource menu, click Add a resource. From the Select a Resource drop-down menu, scroll to LVM. Enter *webfs* for the resource name, and *web01* and *weblv* for the volume group and logical volume names. Click Submit, and click OK in the Update this resource pop-up.

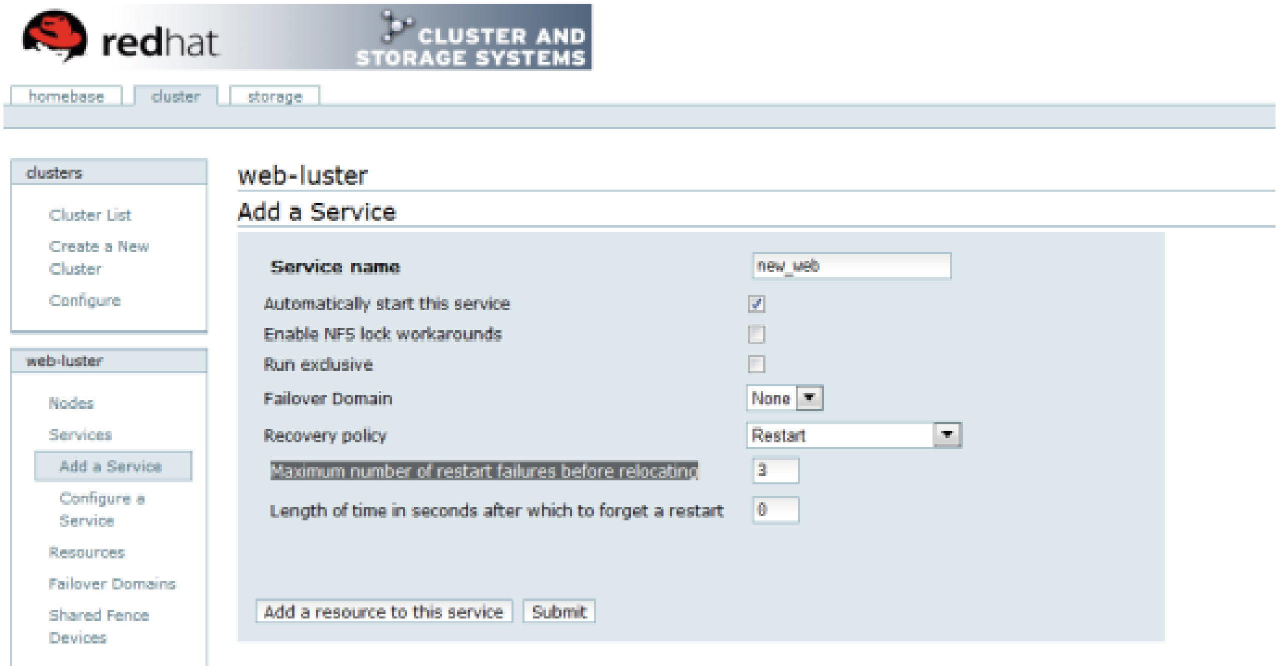


4. Create the file-system resource, named webfss, that resides on the logical volume created in section 7.2.2. From the cluster's Resource menu, click Add a resource. From the Select

a Resource drop-down menu, scroll to File System. Enter *webfss* for the resource name, ext3 for the file system type, */mnt/*web for the mount point, and */dev/mapper/web01-weblv* for the device name. Click Submit, and click OK in the Update this resource pop-up.

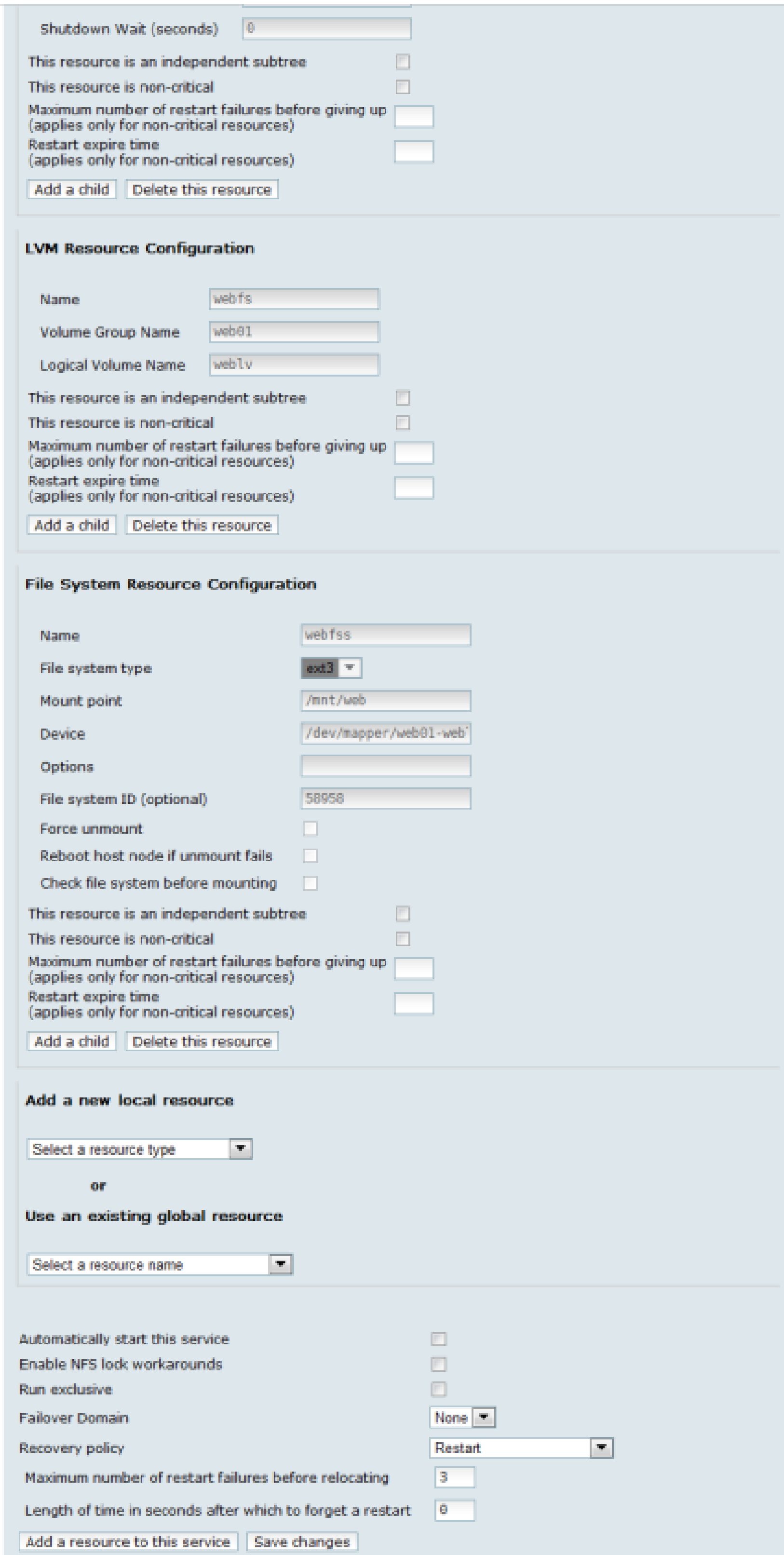


5. Create a cluster service for the web server, named *new_web*. Select the cluster from the menu on the left, click Services, and click Add a service. Enter *new_web* for the service name, check the Automatically start this service box, select Restart as the Recovery policy, and change the Maximum number of restart failures before relocating to an appropriate value, e.g., 3 seconds. Click Submit, and click OK in the Create this service pop-up.



Note: By setting the service's recovery option to *restart*, should the Apache web service stop unexpectedly, the cluster will first try to restart it on the same node. If that fails, the cluster will move the service to the second node, and start it there.

6. Next, assign resources to this service. Select the *new_web* service from the menu at the left.
   a. Click Add a resource to this service. From the Use an existing global resource drop-down menu, select *web (Apache server)*.
   b. Click Add a resource to this service. From the Use an existing global resource drop-down menu, select *10.41.5.106 (IP Address)*.
   c. Click Add a resource to this service. From the Use an existing global resource drop-down menu, select *webfs (Logical Volume Management)*.
   d. Click Add a resource to this service. From the Use an existing global resource drop-down menu, select *webfss (File System).*
   e. Click Save changes, and click OK in the Save changes to this service pop-up.


Shutdown Wait (seconds)
0
This resource is an independent subtree
This resource is non-critical
Maximum number of restart failures before giving up (applies only for non-critical resources)
Restart expire time (applies only for non-critical resources)
Add a child
Delete this resource
LVM Resource Configuration
Name
webfs
Volume Group Name
web01
Logical Volume Name
weblv
This resource is an independent subtree
This resource is non-critical
Maximum number of restart failures before giving up (applies only for non-critical resources)
Restart expire time (applies only for non-critical resources)
Add a child
Delete this resource
File System Resource Configuration
Name
webfss
File system type
ext3
Mount point
/mnt/web
Device
/dev/mapper/web01-web
Options
File system ID (optional)
58958
Force unmount
Reboot host node if unmount fails
Check file system before mounting
This resource is an independent subtree
This resource is non-critical
Maximum number of restart failures before giving up (applies only for non-critical resources)
Restart expire time (applies only for non-critical resources)
Add a child
Delete this resource
Add a new local resource
Select a resource type
or
Use an existing global resource
Select a resource name
Automatically start this service
Enable NFS lock workarounds
Run exclusive
Failover Domain
None
Recovery policy
Restart
Maximum number of restart failures before relocating
3
Length of time in seconds after which to forget a restart
0
Add a resource to this service
Save changes

7.2.4.6 Starting the cluster

Although the web content has not yet been migrated to the new HA cluster, start the cluster services now.

1. At the Services screen, go to the Choose a Task drop-down menu for the service *new_web*, and scroll down to Start this service on *HANode1*. Click Go, and click OK on the pop-up.
2. Repeat step 1 for *HANode2.*

Skip ahead to section 7.2.6.

7.2.5 Migrating Via Command Line to Red Hat Enterprise Linux 6.1 HA

This section of the guide follows the Red Hat Enterprise Linux 6.1 Cluster Administration Guide. Note that the sequence of cluster commands below could be collected into a file as a shell script.

7.2.5.1 Installing Red Hat HA Software

1. Log onto the first server, *HANode1*, as root.
2. Install the HA base software and HA management tools from these two groups of RPMs:

```
# yum groupinstall "High Availability"
# yum groupinstall "High Availability Management"
```

3. Start the secure intra-cluster communication service, ricci:

```
# service ricci start
# chkconfig ricci on
```

4. Assign a password to the *ricci* user account by typing *passwd ricci*.
5. Repeat steps 1-4 for each node.

7.2.5.2 Creating the Two-Node Cluster

Begin by creating the cluster configuration on one node.

1. Log onto the first server, *HANode1*, as root.
2. Create a two-node cluster named *cluster2*. The second command is specific for a two-node cluster and ensures proper functioning with the default quorum rules.

```
# ccs -h HANode1 --createcluster cluster2
# ccs -h HANode1 --setcman two_node=1 expected_votes=1
```

3. Add both nodes to the cluster configuration, as an example, using a simple bash loop:

```
# for host in HANode1 HANode2; do
        ccs -h HANode1 --addnode $host
done
```

7.2.5.3 Configuring I/O Fencing

The cluster design calls for two fencing methods for each node: SCSI-3 PR is the primary method, and iLO is secondary. The shared disk for the SCSI-3 PR fencing will be the shared volume you just created.

The order in which fencing methods are added to a cluster is important. The first method becomes the primary fencing method; the second becomes the backup fencing method.

1. Configure the fencing daemon:

```
# ccs -h HANode1 --setfencedaemon post_fail_delay=3
# ccs -h HANode1 --setfencedaemon post_join_delay=20
```

2. Define the two fencing devices: the first, named *scsi3*, uses the shared */dev/sdb* LUN for SCSI-3 reservations; the second, named *ilo*, uses SSL-protected communications to the (as yet unspecified) server iLO controller to reboot it. We assume both iLOs have the same administrator account *admin* with password *PassworD*.

```
# ccs -h HANode1 --addfencedev scsi3 agent=fence_scsi devices=/dev/sdb
# ccs -h HANode1 --addfencedev ilo agent=fence_ilo_apc \
        login=admin passwd=PassworD
```

3. Add two fencing methods to each node with SCSI-3 PR as primary, and iLO as secondary. The names for the methods are *DiskRes,* and *RebootByIlo*.

```
# for host in HANode1 HANode2; do
        ccs -h HANode1 --addmethod DiskRes      $host
        ccs -h HANode1 --addmethod RebootByIlo $host
done
```

4. Create fencing instances, which also permit individual node configuration:

```
# ccs -h HANode1 --addfenceinst scsi3 HANode1 DiskRes
# ccs -h HANode1 --addfenceinst scsi3 HANode2 DiskRes
# ccs -h HANode1 --addfenceinst ilo HANode1 RebootByIlo ipaddr=10.10.10.127
# ccs -h HANode1 --addfenceinst ilo HANode2 RebootByIlo ipaddr=10.10.10.128
```

5. The SCSI-3 PR fences require an unfencing method

```
# ccs -h HANode1 --addunfence scsi3 HANode1 action=on
# ccs -h HANode1 --addunfence scsi3 HANode2 action=on
```

7.2.5.4 Configuring Failover Domains

For this two-node cluster, configuring failover domains is unnecessary, as the default failover policy suffices: if the active node or a critical component fails, activate the passive node, switch the shared LUN and virtual IP resources to the new active node, and start the web server.

7.2.5.5 Configuring the HA Web Server

The clustered, high-availability Apache web server is a service requiring four cluster resources: the Apache software, the IP address it is bound to, the shared logical volume , and the latter's file system containing its document files. These resources are all part of Red Hat Enterprise Linux HA's default package.

1. Create a cluster service for the web server, named *new_web*. By setting the service's recovery option to *restart*, should the Apache web service stop unexpectedly, the cluster will first try to restart it on the same node. If that fails, the cluster will move the service to the second node, and start it there.

```
# ccs -h HANode1 --addservice new_web recovery=restart
```

2. Add the Apache web server resource, named *web*, specifying its installation location and path to its configuration file.

```
# ccs -h HANode1 --addsubservice new_web apache name=web \
        server_root=/etc/http config_file=/etc/httpd/conf/httpd.conf
```

3. Add the web server's virtual IP address as a resource, and have the link monitored for failure.

```
# ccs -h HANode1 --addsubservice new_web ip address=10.41.5.106 monitor_link=1
```

4. Add the shared logic volume resource, named *webfs*, for the logical volume created in section 7.2.2.

```
# ccs -h HANode1 --addsubservice new_web lvm name=webfs \
        vg_name=web01 lv_name=weblv
```

5. Add the file system resource, named *webfss*, that resides on shared logical volume.

```
# ccs -h HANode1 --addsubservice new_web fs name=webfss \
        mountpoint=/mnt/ device=/dev/mapper/web01-weblv
```

   Note: the default resource-priorities, which determine the order in which resources start, will activate the logical volume on the node *before* the file system is mounted. Alternatively, the file system resource could have been made a child of the logical volume resource for this service to achieve the same start order as well as establishing a logic relation between resources; that is, if the child fails, the parent resource fails too.

7.2.5.6 Starting the cluster

Although the web content has not yet been migrated to the new HA cluster, start the cluster now.

1. Copy the cluster configuration to all nodes, and verify that all nodes have identical configurations:

```
# ccs -h HANode1 --sync --activate
# ccs -h HANode1 --checkconf
```

2. Start the cluster service software on all nodes:

```
# for host in HANode1 HANode2; do
        ssh $host chkconfig cman on \; services cman start \;
                chkconfig rgmanager on \; services rgmanager start
done
```

7.2.6 Stopping VCS Web Server Prior to Copying its Data

At this point, the two migration paths rejoin. The new cluster is running, and the VCS web server can be stopped prior to copying its data to the new cluster.

1. Log onto the active VCS node as root.
2. Stop the Apache web service (denoted here as webservice) on each node:

```
# hagrp –offline webservice –sys node1
# hagrp –offline webservice –sys node2
```

3. Next, stop the remaining VCS cluster service groups in this order on all nodes but this one:

```
# hagrp –offline vrts_vea_cfs_int_cfsmount1 –sys node2
# hagrp –offline cvm –sys node2
```

4. Check that the web server has stopped and that only this node can access its document-root file system:

```
# hastatus –summary
```

```
[root@node1 ~]# hastatus -summary

-- SYSTEM STATE
-- System               State                Frozen

A  node1                RUNNING              0
A  node2                RUNNING              0

-- GROUP STATE
-- Group           System               Probed     AutoDisabled    State

B  ClusterService  node1              Y          N               ONLINE
B  ClusterService  node2              Y          N               OFFLINE
B  cvm             node1              Y          N               ONLINE
B  cvm             node2              Y          N               OFFLINE
B  vrts_vea_cfs_int_cfsmount1 node1   Y          N               ONLINE
B  vrts_vea_cfs_int_cfsmount1 node2   Y          N               OFFLINE
B  webservice      node1              Y          N               OFFLINE
B  webservice      node2              Y          N               OFFLINE
[root@node1 ~]#
```

7.2.7 Copying the Web Server's Configuration Files and Application Files

In Section 5, you examined the main VCS configuration file, determined that the web server's configuration file was */etc/httpd/conf/httpd.conf* and that its application file system was mounted at */mount/web (*which is still mounted on VCS server *node1)*. Copy the web server configuration file for reference, but do not install it.

1. Log onto the active node in the Red Hat Enterprise Linux HA cluster, and copy the application file system from node1 of the VCS cluster.

```
# scp –r node1:/mount/web /mnt/web
```

2. Reboot this server (the active node). The passive node will start the web server with the new web content.

7.2.8 (Optional) Restarting Apache Web Server on the VCS Cluster

If minimizing application down time is important, start the original VCS web server now.

1. Log onto one of the VCS nodes.
2. Start the VCS shared file-system service groups on *node2* in this order:

```
# hagrp –online cvm –sys node2
# hagrp –online vrts_vea_cfs_int_cfsmount1 –sys node2
```

3. Start the Apache web service (denoted here as webservice) on each node:

```
# hagrp –online webservice –sys node2
# hagrp –online webservice –sys node1
```

4. Check that the web server is online on both nodes:

```
# hastatus –summary
```

7.2.9 Starting the HA Web Server

Start the web server as a clustered service:

```
# clusvadm –e ha_web_server
```

7.2.10 Cleaning up

When you determine that you no longer need the VCS cluster, remove the Apache and VCS software from both VCS nodes.

1. Stop the web server and cluster services groups on both nodes as in Section 7.2.5.
2. Stop the VCS cluster on all nodes:

```
# hastop -all -force
```

3. Remove the Apache web server:

```
# yum groupremove "web server”
```

4. Remove these RPMs:

   VRTSamf VRTSaslapm VRTSatClient VRTSatServer VRTScavf VRTSdbed VRTSgab VRTSglm VRTSgms VRTSllt VRTSob VRTSodm VRTSperl VRTSsfmh VRTSspt VRTSvcs VRTSvcsag VRTSvcsdr VRTSvcsea VRTSvlic VRTSvxfen VRTSvxfs VRTSvxvm

The migration is complete.

8 Terms

Term	Meaning
ASIC	Application Specific Integrated Circuit is hardware by iLO for remote management
APC	American Power Conversion Corp.
Cisco MDS	Cisco's series of Multilayer Directors and Fabric Switches
CMAN	The Cluster Manager is responsible managing cluster quorum and keeps track of cluster membership
DRAC	Dell Remote Access Controller is an interface card that provides out-of-band management for the Dell Servers
DM-MPIO	Device Mapper Multipath driver
FC	Fibre Channel
GFS2	Red Hat Global File System 2, a 64-bit, native Linux file system, provides clustered nodes with a consistent view of a shared file system
HA	High Availability
IF-MIB	IETF Interfaces MIB
iLO	HP Integrated Lights Out
IPMI	Intelligent Platform Management Interface
LVM2	A user space toolset that provides logical volume management facilities on Linux
LAN	Local Area Network
NIC	Network Interface Card
NMI Watchdog	Non-maskable Interrupt watchdog relies on specific server hardware to detect system hang
PR	Persistent Reservation is set of SCSI-3 commands that allow multiple systems access to a shared device while preventing others from accessing it.
Qdisk	A disk-based quorum for an RHCS cluster
Qdiskd	A cluster quorum disk daemon provides a mechanism for determining node-fitness in a cluster environment
REST	Representational State Transfer, a web-services API
RIBCL	The Remote Insight Board Command Language is an XML-based dialect used for administration of the iLO
Red Hat Enterprise Linux HA	Red Hat Enterprise Linux High Availability
RSA	Remote Supervisor Adapter provides comprehensive server management on IBM systems
RSB	Fujitsu Siemens's RemoteView Service Board
SAN	storage area network

Term	Meaning
SCSI	Small Computer System Interface is standard for transferring data between computer systems and peripheral devices
SNMP	Simple Network Management Protocol
VCS	Symantec's Veritas Cluster Server
VM	Virtual machine, also known guest
VxVM	Veritas Volume Manager is a logical volume manager
WDT	Watchdog Timer, used for detecting and managing system hangs
WTI	Western Telematic Inc.
XML	Extensible Markup Language is a general-purpose specification for creating custom markup languages

Table 8-1: Terms

9 For More Information

- The administration guide for Red Hat Enterprise Linux 6 with HA Add-On is located at http://docs.redhat.com/docs/en-US/Red_Hat_Enterprise_Linux/6/html/Cluster_Administration/index.html.
- The administration guide for Red Hat Enterprise Linux 5 with HA Add-On is located at http://docs.redhat.com/docs/en-US/Red_Hat_Enterprise_Linux/5/html/Cluster_Administration/index.html.
- Additional information on the Red Hat Enterprise Linux HA Add-On is located at https://fedorahosted.org/cluster/wiki/HomePage. However, take care when referring to documentation on the wiki as formal releases may not completely implement upstream features.
- The Veritas Storage Foundation and High Availability Installation Guide is located at http://www.symantec.com/business/support/index?page=content&id=DOC4252.

10 Appendix

10.1 Sample Scripts

This bash script lists the VCS services and their resources:

```
for i in `hagrp -list | cut -f1 | sort -u `; do
   echo "$i - "; hagrp -resources $i ; echo "" ;
done
```

This bash script lists all nodes on the VCS cluster:

```
hasys -state
```

10.2 Migration Planning Worksheet

Migration Planning: Cluster Parameters	
Symantec VCS Configuration items	**Red Hat Enterprise Linux HA Configuration Items**
Cluster Name ____________________ 1. Node Name____________________ 2. Node Name____________________ 3. Node Name____________________	Cluster Name ____________________ 1. Node Name____________________ 2. Node Name____________________ 3. Node Name____________________
Network Interfaces Heartbeat Network 1. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ 2. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ Client Network 1. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ 2. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ Virtual IP for the Web server IP Address____________________ Netmask____________________	Network Interfaces Heartbeat Network 3. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ 4. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ Client Network 3. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ 4. NIC Name____________________ NIC IP Address____________________ NIC Netmask____________________ Virtual IP for the Web server IP Address____________________ Netmask____________________
Disk Group (LVM or VxVM)____________________ Volume Group Name____________________ Logical Volume Name____________________ Mount Point____________________ Disk Group (LVM or VxVM)____________________ Volume Group Name____________________ Logical Volume Name____________________ Mount Point____________________	Disk Group____________________ Volume Group Name____________________ Logical Volume Name____________________ Mount Point____________________ Disk Group____________________ Volume Group Name____________________ Logical Volume Name____________________ Mount Point____________________
Quorum Disks 1. Device____________________ 2. Device____________________ 3. Device____________________	Quorum Disk Needed? (Y/N)____________________ 1. Device____________________ 2. Device____________________ 3. Device____________________
	iLO IP Addresses

Migration Planning: Cluster Parameters	
	Node 1______________________ Node 2______________________ Node 3______________________
Services (Service Groups) **Service 1** Name______________________ Type______________________ Nodes______________________ Resources Resource 1 Name______________________ Resource Agent______________________ Agent Properties______________________ Critical When Failed?______________________ Resource 2 Name______________________ Resource Agent______________________ Agent Properties______________________ Critical When Failed?______________________ Dependencies Between Resources ______________________ ______________________ **Service 2** Name______________________ Type______________________ Nodes______________________ Resources Resource 1 Name______________________ Resource Agent______________________ Agent Properties______________________ Critical When Failed?______________________ Resource 2 Name______________________ Resource Agent______________________ Agent Properties______________________ Critical When Failed?______________________ Dependencies Between Resources ______________________	Services **Service 1** Name______________________ Type______________________ Nodes______________________ Resources Resource 1 Name______________________ Resource Agent______________________ Agent Properties______________________ Critical When Failed?______________________ Resource 2 Name______________________ Resource Agent______________________ Agent Properties______________________ Critical When Failed?______________________ Dependencies Between Resources ______________________ ______________________

Table 10.2-1: Migration Planning Worksheet

10.3 Configuration of the original VCS cluster

To be complete, this section describes the configuration of the original VCS cluster.

10.3.1 Configuring the VCS Cluster

Perform the following steps to configure the VCS cluster used in the migration:

1. Verify the servers' network and disk configuration.
   node1's network interfaces and IP addresses:
   - eth0: 10.41.5.31
   - eth1: 10.10.10.31

   node2's network interfaces and IP addresses:
   - eth0: 10.41.5.32
   - eth1: 10.10.10.32

Each server has a 50GB boot from SAN disk for the Red Hat OS, LUN0. In addition, map all servers to four shared LUNs:

- Application disk: LUN1, 10GB
- Quorum disk1: LUN11 10GB
- Quorum disk2: LUN12 10GB
- Quorum disk2: LUN13 10GB

2. Type *fdisk –l* from all servers to verify all storage is ready. Use *fdisk* to create partitions on all four shared disks if none are present.

```
[root@bronte1 ~]# fdisk -l

Disk /dev/sda: 53.6 GB, 53687091200 bytes
255 heads, 63 sectors/track, 6527 cylinders
Units = cylinders of 16065 * 512 = 8225280 bytes

   Device Boot      Start         End      Blocks   Id  System
/dev/sda1   *           1          13      104391   83  Linux
/dev/sda2              14        6527    52323705   8e  Linux LVM

Disk /dev/sdb: 107.3 GB, 107374182400 bytes
255 heads, 63 sectors/track, 13054 cylinders
Units = cylinders of 16065 * 512 = 8225280 bytes

   Device Boot      Start         End      Blocks   Id  System

Disk /dev/sdc: 10.7 GB, 10737418240 bytes
64 heads, 32 sectors/track, 10240 cylinders
Units = cylinders of 2048 * 512 = 1048576 bytes

   Device Boot      Start         End      Blocks   Id  System

Disk /dev/sdd: 10.7 GB, 10737418240 bytes
64 heads, 32 sectors/track, 10240 cylinders
Units = cylinders of 2048 * 512 = 1048576 bytes

   Device Boot      Start         End      Blocks   Id  System

Disk /dev/sde: 10.7 GB, 10737418240 bytes
64 heads, 32 sectors/track, 10240 cylinders
Units = cylinders of 2048 * 512 = 1048576 bytes

   Device Boot      Start         End      Blocks   Id  System
```

3. Copy the Install media for Veritas Cluster Services to an NFS share, or mount the CDROM with the mount –o command: *mount -o ro /dev/cdrom /mnt/cdrom*

10.3.2 Performing a Pre-installation Check

Run the installer with the following steps:

1. Type */VRTS_SF_HA_Solutions_5.1_SP1_Red Hat Enterprise Linux /dvd1-redhatlinux/rhel5_x86_64/installer* and press Enter.
2. Select '*P*' and press Enter to launch the pre-installation check.

```
                                   Storage Foundation and High Availability Solutions 5.1 SP1 Install Program

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The Licensed Software and Documentation are deemed to be "commercial computer software" and "commercial computer software documentation" as
defined in FAR Sections 12.212 and DFARS Section 227.7202.

Logs are being written to /var/tmp/installer-201105191222RyL while installer is in progress.

                                   Storage Foundation and High Availability Solutions 5.1 SP1 Install Program

Symantec Product                                   Version Installed  Licensed
==============================================================================
Symantec Licensing Utilities (VRTSvlic) are not installed due to which products and licenses are not discovered.
Use the menu below to continue.


Task Menu:

    P) Perform a Pre-Installation Check     I) Install a Product
    C) Configure an Installed Product       G) Upgrade a Product
    O) Perform a Post-Installation Check    U) Uninstall a Product
    L) License a Product                    S) Start a Product
    D) View Product Descriptions            X) Stop a Product
    R) View Product Requirements            ?) Help

Enter a Task: [P,I,C,G,O,U,L,S,D,X,R,?] P
```

3. Select '*6*' and press Enter for Veritas Storage Foundation Cluster File System/HA (SFCFSHA).

```
                                          Storage Foundation and High Availability Solutions 5.1 SP1 Precheck Program

     1)  Veritas Dynamic Multi-Pathing (DMP)
     2)  Veritas Cluster Server (VCS)
     3)  Veritas Storage Foundation (SF)
     4)  Veritas Storage Foundation and High Availability (SFHA)
     5)  Veritas Storage Foundation Cluster File System (SFCFS)
     6)  Veritas Storage Foundation Cluster File System/HA (SFCFSHA)
     7)  Veritas Storage Foundation for Oracle RAC (SFRAC)
     8)  Veritas Storage Foundation Cluster File System for Oracle RAC (SFCFS RAC)
     9)  Symantec VirtualStore (SVS)
    10)  Symantec Product Authentication Services (AT)
     b)  Back to previous menu

Select a product to perform pre-installation check for: [1-10,b,q] 6
```

4. Enter the 64-bit Red Hat Enterprise Linux 5 system names separated by spaces: type *node1 node2* and press Enter.
5. Set up ssh/rsh communication automatically between the nodes, type *y* and press Enter.
6. Enter the superuser password for system node1: *Password1* and press Enter.

7. Select the communication method, type *1* for SSH, and press Enter.
8. Wait for the installer to verify the servers are ready for install.

```
    Verifying systems: 100%

    Estimated time remaining: 0:00                                                                          8 of 8

    Checking system communication ............................................................................. Done
    Checking release compatibility ............................................................................ Done
    Checking installed product ................................................................................ Done
    Checking prerequisite patches and rpms .................................................................... Done
    Checking platform version ................................................................................. Done
    Checking file space ....................................................................................... Done
    Performing product license checks ......................................................................... Done
    Performing product prechecks .............................................................................. Done

Precheck report completed

System verification checks completed successfully

No issues found in prechecks
```

9. When prompted to install SFCFSHA, type *N* and press Enter.

10.3.3 Installing Veritas Storage Foundation Cluster File System/HA

1. From the install media, type *./installer* and press Enter.
2. From the install menu, press *I*, and press Enter.

```
                                   Storage Foundation and High Availability Solutions 5.1 SP1 Install Program

Symantec Product                                 Version Installed  Licensed
============================================================================
Symantec Licensing Utilities (VRTSvlic) are not installed due to which products and licenses are not discovered.
Use the menu below to continue.


Task Menu:

    P) Perform a Pre-Installation Check     I) Install a Product
    C) Configure an Installed Product       G) Upgrade a Product
    O) Perform a Post-Installation Check    U) Uninstall a Product
    L) License a Product                    S) Start a Product
    D) View Product Descriptions            X) Stop a Product
    R) View Product Requirements            ?) Help

Enter a Task: [P,I,C,G,O,U,L,S,D,X,R,?] I
```

3. Select '*6*' for Veritas Storage Foundation Cluster File System/HA (SFCFSHA), and press Enter.
4. Press *y* to agree with the licensing agreement, and press Enter.
5. Press 2 and Enter to install the recommended SFCFSHA RPMs.
6. Type the Red Hat Enterprise Linux 5 system names to be included in the cluster. Type *node1 node2* and press Enter.
7. Wait for the installer to verify the servers
8. Review the summary of RPMs to be installed, and press Enter to begin.

```
                    Veritas Storage Foundation Cluster File System/HA 5.1 SP1 Install Program
                                             bronte1 bronte2

The following Veritas Storage Foundation Cluster File System/HA rpms will be installed on all systems:

Rpm                 Rpm Description

VRTSvlic            Veritas Licensing
VRTSperl            Veritas Perl 5.10.0 Redistribution
VRTSspt             Veritas Software Support Tools by Symantec
VRTSvxvm            Veritas Volume Manager Binaries
VRTSaslapm          Volume Manager - ASL/APM
VRTSob              Veritas Enterprise Administrator Service by Symantec
VRTSsfmh            Veritas Storage Foundation Managed Host by Symantec
VRTSvxfs            Veritas File System
VRTSatClient        Symantec Product Authentication Service Client
VRTSatServer        Symantec Product Authentication Service
VRTSllt             Veritas Low Latency Transport
VRTSgab             Veritas Group Membership and Atomic Broadcast
VRTSvxfen           Veritas I/O Fencing by Symantec
VRTSamf             Veritas Asynchronous Monitoring Framework by Symantec
VRTSvcs             Veritas Cluster Server
VRTSvcsag           Veritas Cluster Server Bundled Agents by Symantec
VRTSvcsdr           Veritas Cluster Server Disk Reservation Modules
VRTSvcsea           Veritas Cluster Server Enterprise Agents by Symantec
VRTSdbed            Veritas Storage Foundation Databases
VRTSglm             Veritas Group Lock Manager
VRTScavf            Veritas Cluster Server Agents for Cluster File System
VRTSgms             Veritas Group Messaging Services
VRTSodm             Veritas Oracle Disk Manager

Press [Enter] to continue:
```

9. Press 2 and Enter to enable keyless licensing and complete system licensing.
10. Press *y* and Enter to enable the Veritas Volume Replicator.
11. Press *y* and Enter to enable the global cluster option.
12. Press *y* to configure SFCFSHA on all nodes in the cluster.

10.3.4 Configuring I/O Fencing-Network

1. After the initial setup, configure I/O fencing. Press *y* to begin in enabled mode.

```
                 Veritas Storage Foundation Cluster File System/HA 5.1 SP1 Configure Program
                                          bronte1 bronte2

Logs are being written to /var/tmp/installer-201105191422KoF while installer is in progress

    Verifying systems: 100%

    Estimated time remaining: 0:00                                                           5 of 5

    Checking system communication ........................................................ Done
    Checking release compatibility ....................................................... Done
    Checking installed product ........................................................... Done
    Checking platform version ............................................................ Done
    Performing product prechecks ......................................................... Done

System verification checks completed successfully

I/O Fencing

It needs to be determined at this time if you plan to configure I/O Fencing in enabled or disabled mode, as well as help in determining the
number of network interconnects (NICS) required on your systems. If you configure I/O Fencing in enabled mode, only a single NIC is required,
though at least two are recommended.

A split brain can occur if servers within the cluster become unable to communicate for any number of reasons. If I/O Fencing is not enabled,
you run the risk of data corruption should a split brain occur. Therefore, to avoid data corruption due to split brain in CFS environments,
I/O Fencing has to be enabled.

If you do not enable I/O Fencing, you do so at your own risk

See the Administrator's Guide for more information on I/O Fencing

Do you want to configure I/O Fencing in enabled mode? [y,n,q,?] (y) y
```

2. Press Enter at the instructions screen to continue.
3. Enter *VCS-1* for the unique cluster name, and press Enter.
4. Enter 1 for the unique cluster ID number, and press Enter.
5. Press 3 to automatically detect configuration for LLT over Ethernet.
6. Verify the cluster heartbeat information. The system will assign *eth1* as the primary heartbeat, and *eth0* as a low-priority heartbeat.

```
                 Veritas Storage Foundation Cluster File System/HA 5.1 SP1 Configure Program
                                          bronte1 bronte2

Cluster information verification:

        Cluster Name:      VCS-1
        Cluster ID Number: 1
        Private Heartbeat NICs for bronte1:
                link1=eth1
        Low-Priority Heartbeat NIC for bronte1:
                link-lowpri1=eth0
        Private Heartbeat NICs for bronte2:
                link1=eth1
        Low-Priority Heartbeat NIC for bronte2:
                link-lowpri1=eth0

Is this information correct? [y,n,q,b,?] (y) y
```

7. Press *y* and Enter to set up a virtual IP address to allow the cluster to be managed by a Java GUI.
8. Type *eth0* and press Enter to use the NIC for the virtual IP.
9. Press *y* to use *eth0* for all nodes.
10. Enter the virtual IP address: *10.41.2.40* and press Enter.
11. Enter the subnet mask for the virtual IP address: *255.255.248.0* and press Enter.
12. Verify the Virtual IP information, and press *y* to continue.

```
Veritas Storage Foundation Cluster File System/HA 5.1 SP1 Configure Program
bronte1

Cluster Virtual IP verification:

        NIC: eth0
        IP: 10.41.2.40
        NetMask: 255.255.248.0

Is this information correct? [y,n,q] (y)
```

13. Press *n* to skip setting up Symantec Security Services.
14. Keep the default user ID and password (admin/password) press *n*, and press Enter.
15. Press *n* to skip adding additional users.
16. Press *y* to accept the default admin user for VCS-1.
17. Press *n* to skip SMTP notification.
18. Press *n* to skip global cluster option.
19. Press *y* to stop and restart SFCFSHA processes.
20. When prompted for fencing configuration, press *q* to quit.

```
Veritas Storage Foundation Cluster File System/HA 5.1 SP1 Configure Program
bronte1 bronte2

Fencing configuration
     1)  Configure CP client based fencing
     2)  Configure disk based fencing

Select the fencing mechanism to be configured in this Application Cluster: [1-2,q] q
```

10.3.5 Initializing the Quorum LUNs

1. Type *vxdisk -e list* to view the disks available to VCS.

```
[root@bronte1 ~]# vxdisk -e list
DEVICE       TYPE          DISK        GROUP        STATUS               OS_NATIVE_NAME   ATTR
fas9600_0    auto:none     -            -           online invalid       sdc              tp
fas9600_1    auto:none     -            -           online invalid       sdd              tp
fas9600_2    auto:none     -            -           online invalid       sdb              tp
fas9600_3    auto:none     -            -           online invalid       sde              tp
fas9600_4    auto:none     -            -           online invalid       sda              tp
```

2. Make note of the DEVICE names for sdc, sdd and sde (i.e., fas9600_0, fas9600_1, and fas9600_3), and run these commands:

   # /opt/VRTS/bin/vxdisksetup –if fas9600_0
   # /opt/VRTS/bin/vxdisksetup –if fas9600_1
   # /opt/VRTS/bin/vxdisksetup –if fas9600_3
   # vxdg init vxfencoorddg disk_0 disk_1 disk_2
   # vxdg -g vxfencoorddg set coordinator=on

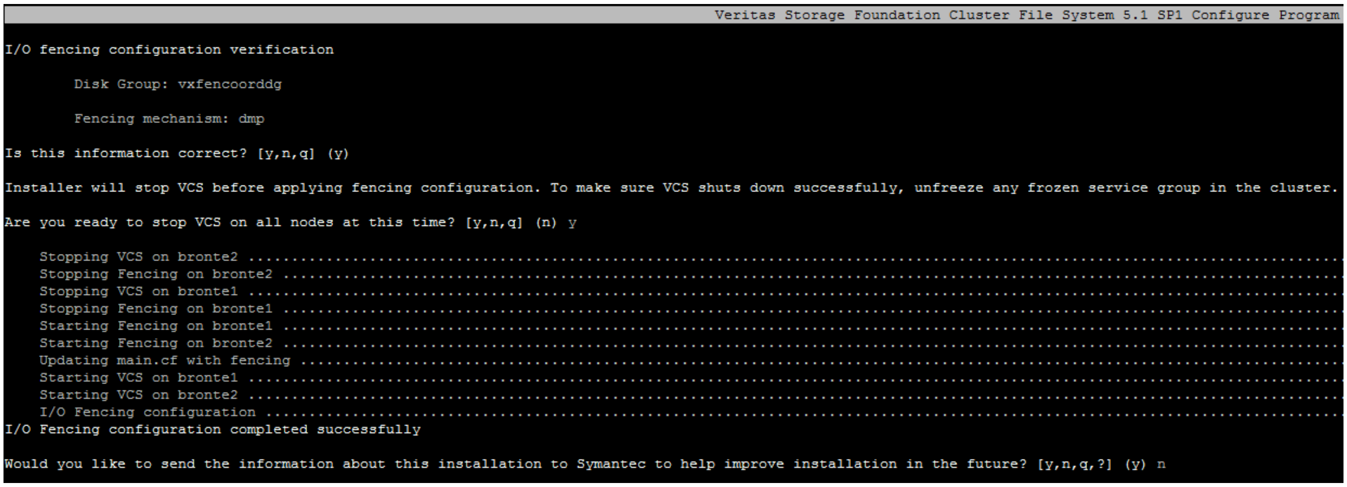
10.3.6 I/O fencing-setup

1. Start the fencing wizard:

   # /opt/VRTS/install/installsfcfs –fencing

2. Enter y to configure fencing on the VCS cluster
3. Press 2 to configure disk based fencing
4. Enter 2 for disk based fencing
5. Enter y for SCSI3 OR enabled disks
6. Enter 2 for existing disk group

7. Enter 1 for vxfencoorddg
8. Enter y to confirm that the disk group is a supported configuration
9. Enter dmp for the fencing mechanism (multipathing)
10. Enter y to confirm the I/O fencing configuration.
11. When prompted to stop VCS on all nodes, ensure all nodes are up, and press y.



12. To ensure fenced nodes reboot after kernel panic, edit the */etc/sysctl.conf* file, and add the following lines:

 # force a reboot after 60 seconds
kernel.panic = 60

```
# Kernel sysctl configuration file for Red Hat Linux
#
# For binary values, 0 is disabled, 1 is enabled.  See sysctl(8) and
# sysctl.conf(5) for more details.

# Controls IP packet forwarding
net.ipv4.ip_forward = 0

# Controls source route verification
net.ipv4.conf.default.rp_filter = 1

# Do not accept source routing
net.ipv4.conf.default.accept_source_route = 0

# Controls the System Request debugging functionality of the kernel
kernel.sysrq = 0

# Controls whether core dumps will append the PID to the core filename
# Useful for debugging multi-threaded applications
kernel.core_uses_pid = 1

# Controls the use of TCP syncookies
net.ipv4.tcp_syncookies = 1

# Controls the maximum size of a message, in bytes
kernel.msgmnb = 65536

# Controls the default maxmimum size of a mesage queue
kernel.msgmax = 65536

# Controls the maximum shared segment size, in bytes
kernel.shmmax = 68719476736

# Controls the maximum number of shared memory segments, in pages
kernel.shmall = 4294967296

#force a reboot after 60 seconds
kernel.panic = 60
```

10.3.7 Creating Shared Storage

1. From the master node run *vxdisk –e list* and note the device name for sdb (in this case sdb=fas9600_4).
2. Initialize the shared storage disk by typing *vxdisksetup –if fas9600_4*
3. Add the shared storage to a disk group (vxvol1) by typing *vxdg init vxvol1 fas9600_4*
4. Note the largest disk size of the 100GB disk by typing *vxassist –g vxvol1 maxsize*
5. Use the *maxsize* number from the above step to create a new volume called vol1 using the maximum available space by typing *vxassist make -g vxvol1 vol1 209532928*
6. Create a cluster filesystem on vol1 by typing *mkfs -t vxfs /dev/vx/dsk/vxvol1/vol1*
7. Deport the new disk group and reimport as shared:

   # vxdg deport vxvol1
   # vxdg -s import vxvol1

8. Grant shared access to the hosts in the cluster by typing *cfsdgadm add vxvol1 all=sw*
9. Create a mount point for the share by typing *mkdir /mount/vol*
10. Create the same mount point on the other nodes.

11. Register the new volume with all nodes and mount it on all nodes:

```
# cfsmntadm add vxvol1 vol1 /mount/vol1 all=cluster
# cfsmount /mount/vol1
```

10.3.8 Installing the Web Server

1. Log into the master node.
2. Install the Apache web server by typing *yum groupinstall "web server"*
3. Edit */etc/httpd/conf/httpd.conf*, and find the section called *DocumentRoot* and change the path to */mount/vol1/www/html*
4. Copy the default web content to the shared storage by typing:

```
# mkdir –p /mount/vol1/www
# cp –rp /var/www/html /mount/vol1/www
```

5. Repeat steps 1-4 for all nodes in the cluster.

10.3.9 Creating the Apache Web Server Group

1. Create a service group called 'webservice' and add both nodes.

```
# hagrp -add webservice
# hagrp -modify webservice SystemList bronte2 0 bronte1 1
# hagrp -modify webservice AutoStartList bronte2 bronte1
# hagrp -modify webservice Parallel 0
```

2. Create an IP-address resource named WEB-IP in the webservice service group.

```
# hares -add WEB-IP IP webservice
# hares -modify WEB-IP Critical 1
# hares -modify WEB-IP PrefixLen 1000
# hares -modify WEB-IP Device eth0
# hares -modify WEB-IP Address 10.41.5.41
# hares -modify WEB-IP NetMask 255.255.248.0
# hares -modify WEB-IP Options
# hares -modify WEB-IP IPOptions
# hares -modify WEB-IP IPRouteOptions 10.41.0.1
# hares -modify WEB-IP Enabled 1
```

3. Create and add an Apache web server resource named apache to the webservice service group.

```
# hares -add apache Apache webservice
# hares -modify apache Critical 1
# hares -modify apache ResLogLevel INFO
# hares -modify apache Port 80
# hares -modify apache SecondLevelMonitor 0
# hares -modify apache SecondLevelTimeout 30
# hares -modify apache EnableSSL 0
# hares -modify apache httpdDir /usr/bin
# hares -modify apache SharedObjDir
# hares -modify apache EnvFile
# hares -modify apache PidFile
# hares -modify apache HostName
# hares -modify apache User
# hares -modify apache ConfigFile /etc/httpd/conf/httpd.conf
# hares -modify apache DirectiveAfter -delete -keys
```

hares -modify apache DirectiveBefore -delete -keys
hares -modify apache Enabled 1
```

4. Modify the apache service retry behavior so that the system attempts to restart a failed apache service three times before initiating a failover.

```
# hagrp -modify webservice OnlineRetryLimit 3
```

5. Link the Apache service to be dependent on the web-IP resource.

```
# hares -link WEB-IP apache
```

6. Link, as a dependent, the webservice service group to the ClusterService and the cluster volume manager.

```
# hagrp -link ClusterService webservice online local firm
# hagrp -link cvm webservice online local firm
```
```